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Towards Coordinated Dual-arm Snap-fit Assembly Skill for Delicate Applications

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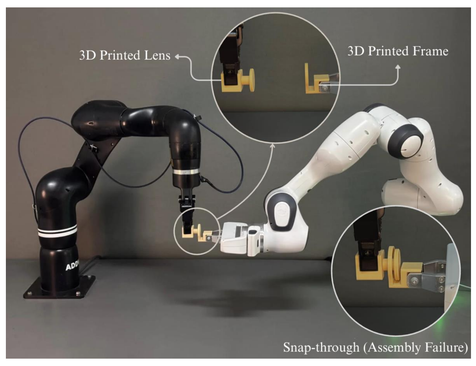

Fig. 1: Bimanual snap-fit task performed under the proposed framework.

Introduction

Snap-fit mechanisms are widely used in **industrial assembly** due to their simplicity, speed, and cost-effectiveness. However, automating these insertions with bimanual robots for delicate parts like electronics or eyewear poses significant challenges, such as **tight tolerances**, brief but **forceful engagement events**, and the risk of **overshoot-induced part damage** or assembly failure. In this work, we introduce a bimanual manipulation framework that integrates both capabilities into a **unified skill for snap-fit assembly**.

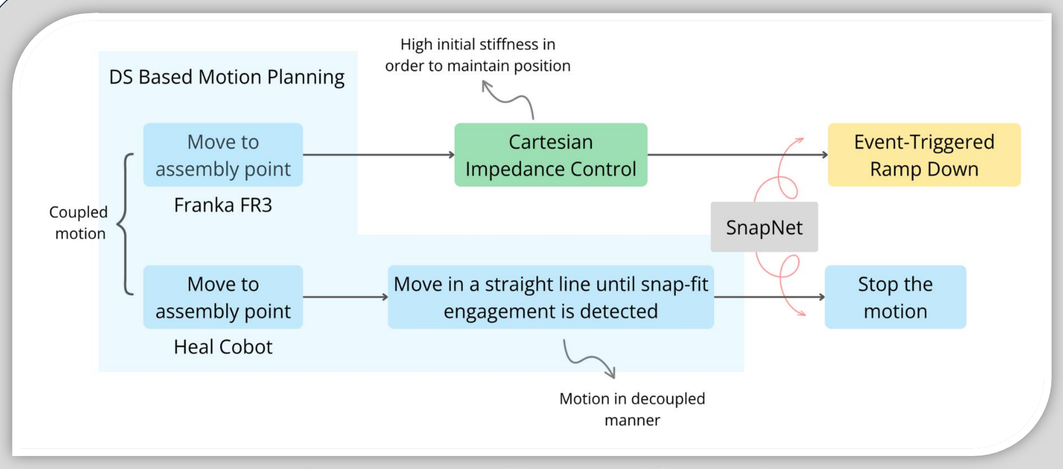

Fig. 2: Overview of the proposed framework.

Our Approach

We combine **Snap-Net (1)** for proprioceptive snap detection with a **coupled DS (2)** for dual-arm coordination, then **trigger event-based VIC (3)** to **exponentially reduce stiffness** along the insertion axis, ensuring safety.

1. Snap-Net Architecture

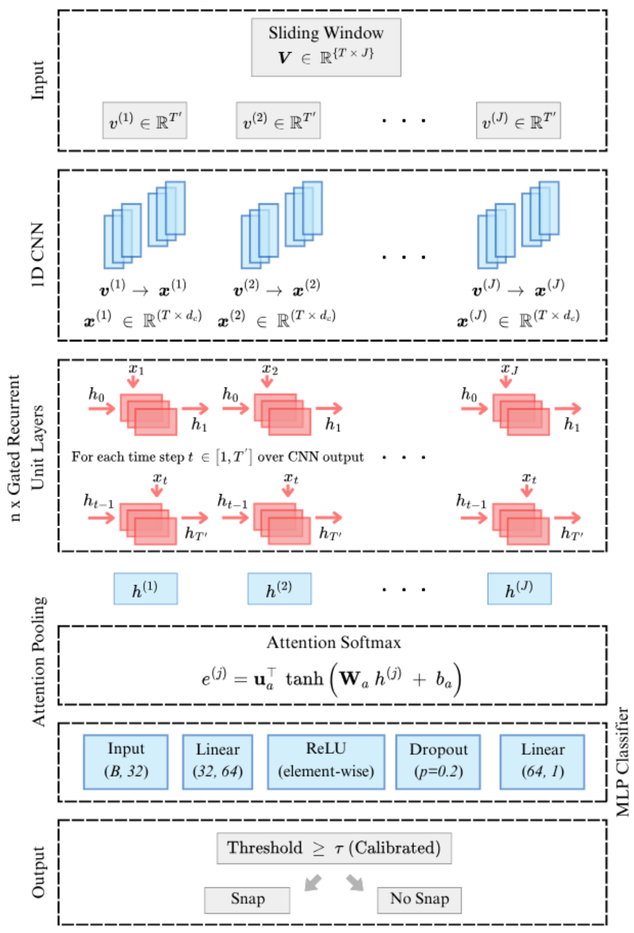

Fig. 3: CNN-GRU based snap detection architecture

2. Multi-Arm Coordination Framework

Coupled dynamical systems ensure smooth multi-arm motion, with phase gating.

$$\dot{z}_i = -k_1 \frac{z_i - z_i^d}{(|z_i - z_i^d| + \varepsilon)^{1-\eta}} - \kappa(\theta)\, k_2 \frac{z_i - z^c}{(|z_i - z^c| + \varepsilon)^{1-\eta}}$$

$$\dot{x}_i = -\alpha_i(E_i) e_i$$

$\kappa = 1 \rightarrow$ synchronized pick/transport.
$\kappa = 0 \rightarrow$ decoupled insertion.
Convergence guaranteed via Lyapunov-based analysis.

3. Event-Triggered Impedance Adaptation

When Snap-Net detects engagement, stiffness is exponentially ramped down by the following time-varying rule.

$$K(t) = K_{\text{low}}\left(1 - e^{-\lambda(t-t_s)}\right) + K_{\text{high}} e^{-\lambda(t-t_s)},$$

$$D(t) = \tfrac{1}{2}\sqrt{K(t)}$$

Experimental Setup

Experiments on Franka FR3 and Addverb Heal performing lens-in-frame and related snap-fits. We stream **joint velocities (100 Hz)** and train on labeled insertions. **Baselines: position and fixed impedance**.

Results

- Snap-Net (CNN–GRU–attention), trained solely on joint velocities, **detects snap-fit engagement in <50ms** and generalizes across robotic platforms.
- Event-triggered compliance achieves ~ **90% task success and lower peak contact forces** than fixed-impedance baselines, outperforming position control.

Snap-Fit Part	FR3	Heal
Marker Pen Cap	14/15 (93.3%)	13/15 (86.7%)
Highlighter Cap	15/15 (100.0%)	14/15 (93.3%)
Bottle Lid Clip	15/15 (100.0%)	—
USB-C/Lightning Connector	13/15 (86.7%)	8/15 (53.3%)
3D-printed Lens & Frame	15/15 (100.0%)	12/15 (80.0%)
Emergency-Stop Button	—	15/15 (100.0%)
Custom 3D-printed Part	—	10/15 (66.7%)

Table 1: Snap-Net Detection Successes (15 trials / condition)

Control Mode	Success Rate	F (max) (N)
Position Control	6/15 (40%)	25.1 ± 3.8
Fixed Impedance	11/15 (73%)	22.4 ± 4.2
Event-Triggered VIC (Proposed)	15/15 (100%)	16.3 ± 3.1

Table 2: Assembly Success & Peak Forces (Lens-Frame Task)

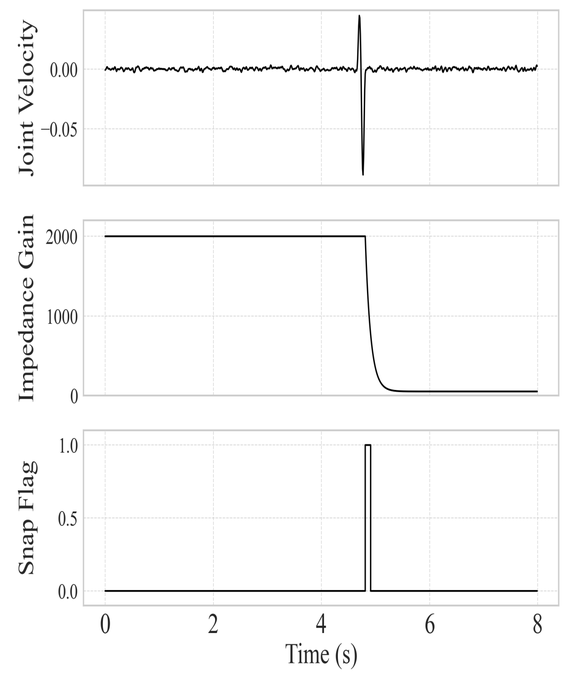

Fig. 4: Top: Joint velocity transient indicating snap event. Middle: Exponential decay of impedance gain triggered post-engagement. Bottom: Binary snap detection flag: initiates compliance modulation.

Project Webpage : https://shr-eyas.github.io/SNAP/

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